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Shutdowns & Overhauls Management

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Asset Management, Glencore Copper
18 November 2021, Version 1.0

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Preface

The Asset Management Framework, Policy and Manuals are intended to provide leading Asset Management practices when applied and will be continually improved for the benefit of Glencore Copper.

The real value of sharing the knowledge gained from the use of these documents will come from continued discussion of the issues and rationale behind each document rather than by reading and interpreting them as individuals.

Readers are encouraged to consider these against our values around Safety, Responsibility, Simplicity, Integrity, Openness and Entrepreneurialism and how they reflect our Purpose – to responsibly source the commodities that advance everyday life. This document lays out the minimum standard for Glencore Copper and compliance to this manual will demonstrate a maturity value equal to Planned/Deployed (3) on the Glencore Copper Asset Management Maturity Scale.

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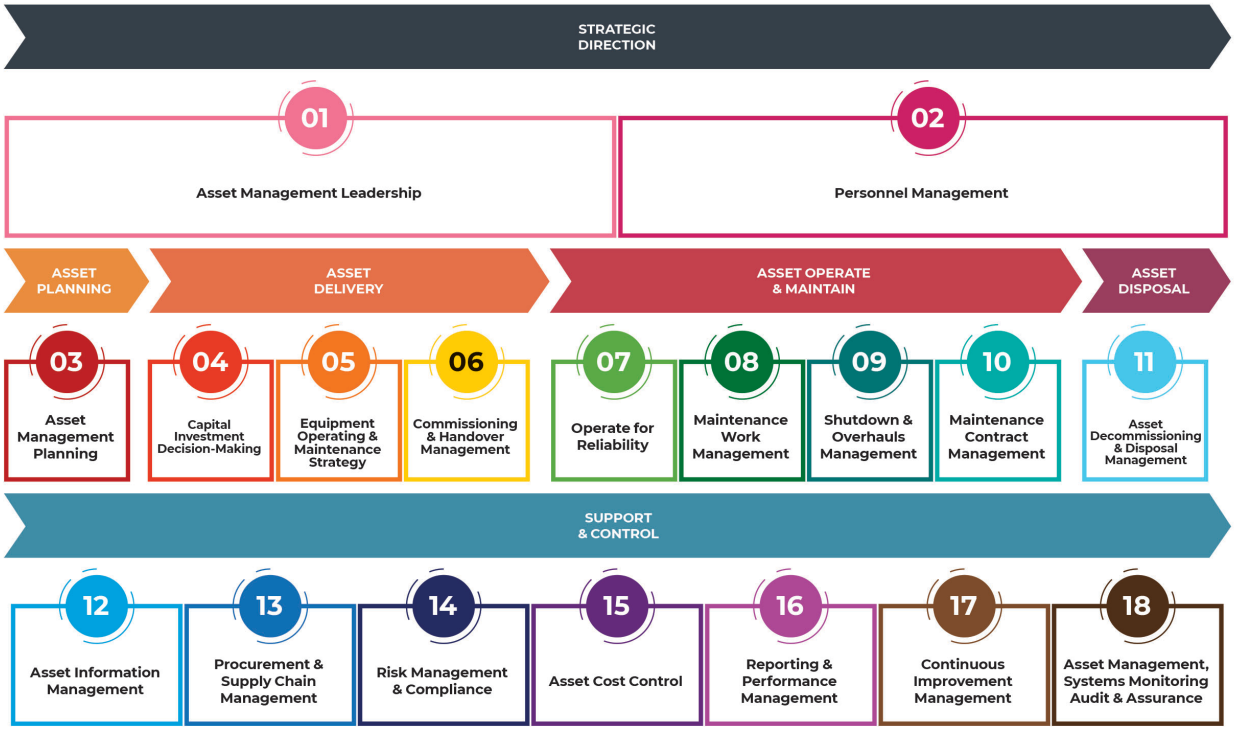
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1. Introduction

The Asset Management Policy and Manuals define how Glencore Copper manages the whole of life of an asset to ensure optimal value is extracted by ensuring assets are capable of meeting business and stakeholder objectives.

Figure 1-1 describes the high-level Asset Management Framework, which enables Glencore Copper to deliver its Asset Management purpose and vision.

FIGURE 1-1 ASSET MANAGEMENT FRAMEWORK



1.1 Shutdown

Shutdowns are major events in terms of production impact, maintenance expenditure, and ensuring correct maintenance and projects are undertaken to improve the equipment reliability and plant availability.

Shutdowns occur over longer time periods than day-to-day maintenance, contain interaction between multiple tasks and have significant costs. Additional project management competencies and processes shall be in place to achieve successful outcomes.

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2. Objectives

The objective of this manual for Shutdowns and Overhauls is to define the performance requirements to enable shutdown and overhaul work to be planned, scheduled, executed, closed out and improved. It also defines how shutdowns and overhauls are to be performed safely, efficiently and with precision and to ensure ramp down and up is done in a timely manner to maximise production. This ensures work is conducted the first time, every time, at the right time and cost, using effective shutdown and overhaul processes together with work management practices.

2.1 Accountabilities

Dedicated resources must be assigned, and capabilities developed within each Operation to deliver Shutdown and Overhaul Management processes to enable effective delivery of the Asset Management objectives.

- Assign accountability for each or all of the following:
  - Shutdown and Overhaul Management identification, planning and execution.
  - Shutdown and Overhaul Management strategies defined and documented.
  - Compliance to the Shutdown and Overhaul Management schedule
- Define and document Shutdown and Overhaul Management competencies required for planning, scheduling and execution of Shutdown and Overhaul Management processes.
- Plan deliver and record Shutdown and Overhaul Management training to support the embedment of required Shutdown and Overhaul system and process competence and knowledge.
- Provide sufficient user access points to operational staff to enable the entry of Shutdown and Overhaul Management information into the Computerised Maintenance Management System (CMMS) and other systems as applicable.

2.2 Scope

This manual is to be read in conjunction with the following manuals:

- E01 Asset Management Leadership
- E04 Capital Investment Decision-Making
- E06 Commissioning and Handover
- E08 Maintenance Work Management
- E10 Maintenance Contracting and Outsourcing

The scope of Shutdown and Overhaul Management covers the following:

- Major Events Calendar, Planned Component Replacement program and Servicing tasks.
- All shutdown and overhaul work activities to be entered into the CMMS.
- Shutdown and Overhaul Management activities with any of the following criteria:
  - involve shutdown and overhaul work on Glencore Copper owned and operated equipment that is safety critical or essential to production.
  - are triggered by equipment defined maintenance strategies and corrective work requirements.
  - impact the reliability of a maintainable item in the CMMS or are associated with defect elimination.
  - involve the logging of history.
  - involve statutory or legislative requirements, including those which are associated with third party or contractor owned equipment where required.
  - require plant spares to be requisitioned from a work order.
  - form part of the sustaining Capital program
  - form part of the asset modification or upgrade program

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3. Definitions

Computer Maintenance Management System (CMMS) - is a business system and software that helps organisations digitize, automate and optimize their maintenance operations. A CMMS software provides a single source of truth for all operation and maintenance information, helping users manage work orders, track inventory, schedule maintenance operations and more from one centralized, real-time system.

Critical Path - The sequence of Work Orders within a shutdown schedule for which there is no float from the commencement of the shutdown to the completion of all works.

Emergent Work - New work determined as a result of conducting scheduled work within a shutdown. Alternatively, new work found as a result of PM inspections that has occurred after the shutdown plan has been locked or found as a result of an unplanned equipment, plant or component failure that is added to the shutdown post shutdown risk assessment. This work is typically budgeted for in contingency funds and there is sufficient labour available to do the work providing there is float allowed into the shutdown schedule.

Major Events Calendar (MEC)– a calendar of all the major activities to be undertaken in a predefined planning period, derived from, Life of Asset Plan, Strategic Asset Management Plan and annual Budget formulation. Major activities include mine planning, acquisitions, shutdowns, overhauls, capital works, and disposal.

Planned Component Replacement (PCR) – the practise of replacing components before they fail and derived from the Equipment Operating and Maintenance strategy process contained within E05 of the Glencore Cu Asset Management Framework.

Shutdowns and Overhauls – A planned stoppage of the plant or equipment which allows a mix of project and major maintenance work to be undertaken according to a well-defined schedule, which coordinates a large number of resources to work in the same vicinity to each other. It is critical that the return to service from the shutdown complies with the production schedule. The shutdown duration is from full capacity prior to ramp down to full capacity after start up.

Work Breakdown Structure (WBS) - is a deliverable-oriented breakdown of a project into smaller components. A work breakdown structure is a key project deliverable that organizes the team's work into manageable sections. The Project Management Body of Knowledge (PMBOK 5) defines the work-breakdown structure as a "hierarchical decomposition of the total scope of work to be carried out by the project team to accomplish the project objectives and create the required deliverables."

4. Approach

The approach for Shutdown and Overhaul Management is described through:

- Process Flow - pictorially illustrates the key Shutdown and Overhaul Management activities, refer to Figures 4-1 through Figure 4-7.
- Process Description – defines in more detail the key Shutdown and Overhaul Management activities, refer to Section 4.2.
- Process RACI Matrix - indicates for each activity, the positions who are Responsible, Accountable, who are Consulted prior to final decision/action, and those who are Informed of any outcomes and actions, refer to Section 4.

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Shutdowns & Overhauls Management

4.1 Process Flow – Level 2

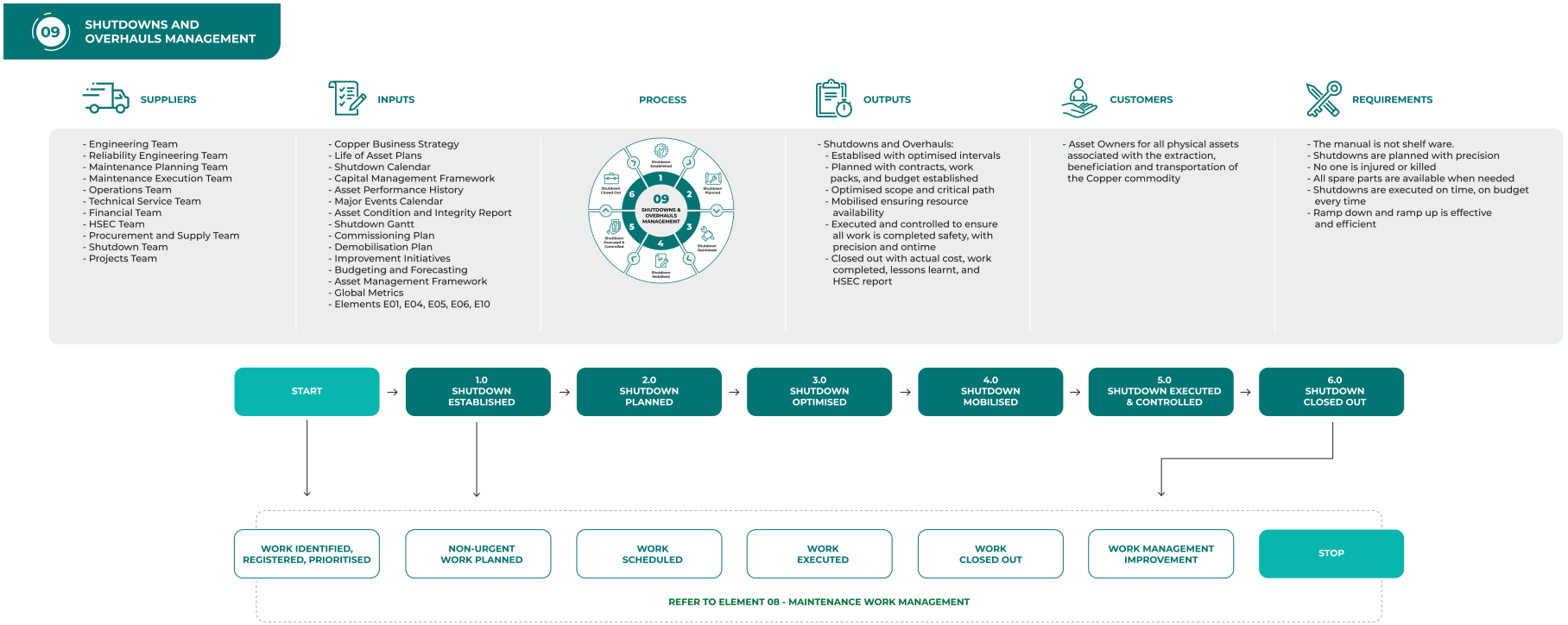


FIGURE 4-1 SHUTDOWNS AND OVERHAULS MANAGEMENT PROCESS

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4.1.1 Shutdown Established – Level 3

Planned tasks identified in the approved annual business, major events and budget plan are assembled into the equipment shutdown schedule and defined by a critical path activity.

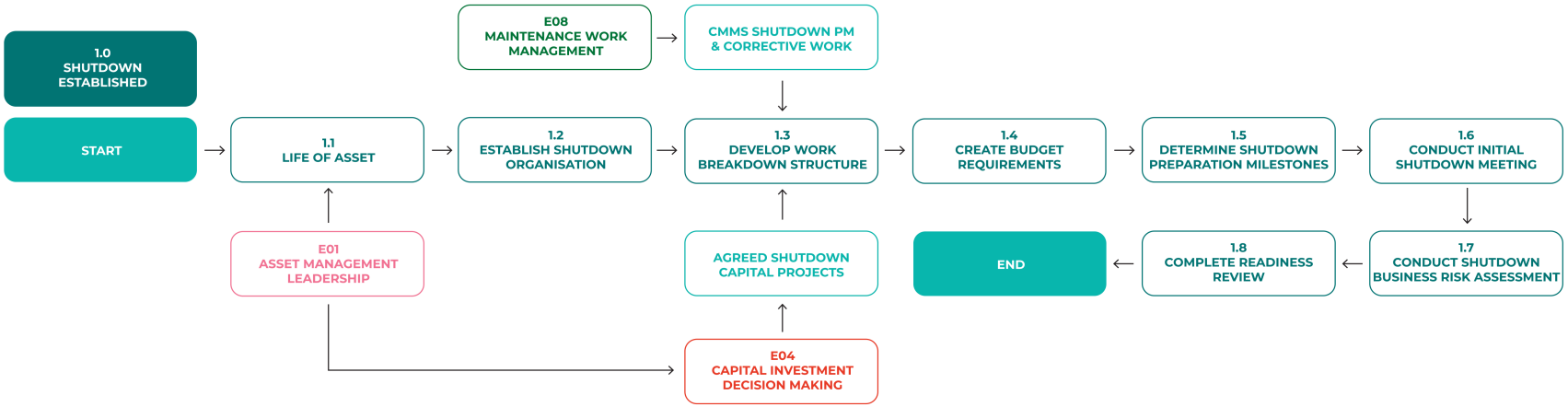


FIGURE 4-2 SHUTDOWN ESTABLISHED PROCESS

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TABLE 4-1 SHUTDOWN ESTABLISHED PROCESS DESCRIPTION

PROCESS STEP	PROCESS DESCRIPTION	DESCRIPTION
1.1	Life of Asset Plan	Refer to Element E01 Asset Management Leadership. The Life of Asset Plan determines the Major Events Calendar (MEC), Planned Component Replacement (PCR) program, budget physicals and expenditure and shall be reviewed to determine the timing of the shutdown. Each site shall start planning for major shutdowns and overhauls earlier enough to ensure all long lead time items are procured and onsite well in advance of the execution phase.
1.2	Establish Shutdown Organisation	Determine the members of the Shutdown Steering and Working groups, this will generally include managers, superintendents, engineers, planners, coordinators and contractors as required. Steering group instigate the Shutdown and establish the shutdown including: • Shutdown organisation structure • Roles and responsibilities • Shutdown duration • Contracts established. • Shutdown Risk assessment • Emergent Work Control • Budget Approval and Cost Management
1.3	Develop Work Breakdown Structure (WBS)	The Work Breakdown Structure (WBS) is a hierarchy of shutdown activities that breaks the 'shutdown' down, in a logical fashion, to work package level. The logic should be such that a work package can be easily located in the structure by drilling down from the top level of the 'shutdown'. The initial Shutdown WBS to be derived from CMMS generated shutdown PM tasks (refer to E05 Equipment Operating and Maintenance Strategy), corrective tasks and agreed shutdown Capital Projects (refer to E04 Capital Investment Decision-Making) and is represented in the shutdown scheduling tool as the baseline schedule. A Shutdown Gantt chart is to be established containing the key activities and critical path. To assist in the development of the WBS the following should be considered: • The assets deliverables, (performance criteria), • Complete lifecycle of the asset, • Current asset life • Current condition of the asset • Innovative or new technologies to be installed in the asset, • Engineering improvements to be installed in the asset, • Industry, OEM or other standards, updates or requirements • Task dependencies • Key milestones
1.4	Create Budget Requirements	Develop Shutdown work orders with shutdown revision number, information to include as a minimum: • Task duration • Task description, a procedure may need to be developed based on the complexity of the job. • Resources – human, tools, equipment, spares, consumables, drawings, risk assessment etc. • Total work order planned cost

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1.5	Determine Shutdown Preparation Milestones	All shutdown tasks to be assessed to determine the activities required leading up to the shutdown. Shutdown milestones shall be established for all stakeholders and should consider: • When each team member should commence planning activities • Determine how long they are required • Duration for each activity or engagement The shutdown meeting schedule shall be established and issued to stakeholders, defining each toll gate session, date, duration and information protocols.
1.6	Conduct Initial Shutdown Meeting	Stakeholders present a detailed overview of their shutdown tasks. The Shutdown Planner/Specialist shall provide: • Shutdown objectives and overview • Shutdown organisation structure • Key identifiable risks • Identification of additional jobs • Safety Strategy
1.7	Conduct Shutdown Business Risk Assessment	All shutdown tasks to be collated into the shutdown risk assessment with an objective to: • Assess the risk of not doing a job. • Assess the risk of not being able to complete a job. • Assess task criticality. • Assess task priority. • Assess each task cost
1.8	Complete Readiness Review	All Shutdown information is presented to the relevant Manager as per the delegation of authority for sign off. The following activities shall be complete at this stage: • Shutdown Risk Assessment signed off. • Go/no go to proceed to next shutdown phase. • Tasks assessed and approved or rejected. • Operations are engaged and communications provided to the Operations Planner regarding shutdown durations and impacts. • Shutdown extensions compared to budget shall require shutdown analysis and approved by the respective Manager. • Safety action plan.

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4.1.2 Shutdown Planned – Level 3

Shutdown Planned takes the established shutdown and defines each shutdown task into a consolidated plan. In this phase task details are defined, and methodologies approved.

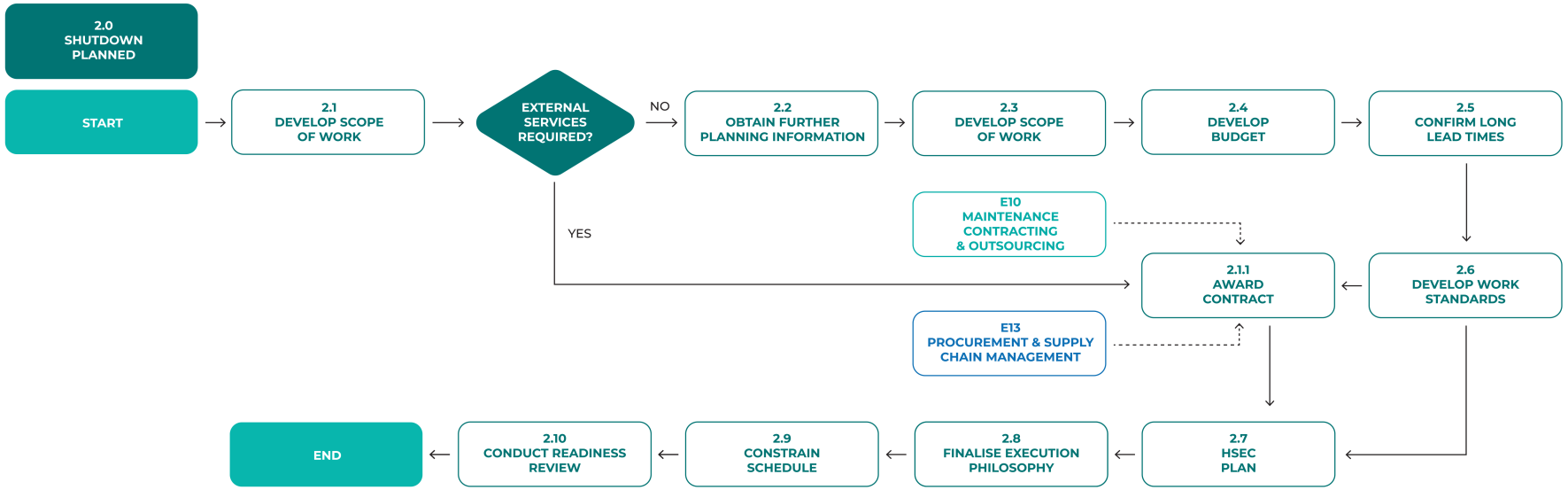


FIGURE 4-3 SHUTDOWN PLANNED PROCESS

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TABLE 4-2 SHUTDOWN PLANNED PROCESS DESCRIPTION

PROCESS STEP	PROCESS DESCRIPTION	DESCRIPTION
2.1	Develop Scope of Work	Scope of work for the shutdown is defined and assembled in the shutdown folder or equivalent. A detailed shutdown schedule together with the shutdown workbook or equivalent is completed. Outcomes from the Business Risk Assessment are utilised to develop a scope for each task. Key items are reviewed such as: • Resources available to complete each task. • Parts and material identified. • Commercial engagements are drafted. • Asset back logs assessed. • Safety Strategy • Review previous shutdown lessons learnt. • Review Root Cause Failure Analysis for any corrective actions.
2.1.1	Are External Services Required?	If shutdown external services/contractor are required then refer to E13 Procurement and Supply Chain Management for selection and engagement requirements and E10 Maintenance Contracting and Outsourcing, then 2.7, else go to 2.2. Where practical contractors that have been engaged in shutdown activities before on site should be targeted. This will ensure consistency of delivery and reduction in induction time and expense.
2.2	Obtain Further Planning Information	The Shutdown Planner/Specialist is to engage the stakeholders and obtain further planning information, such as: • What is the job definition? • Who must do the job steps – what skills are required? • How many people are required? • What parts, materials, or supplies will be needed? • What layout do we need? • Is any support equipment required – e.g., scaffolding etc.? • How long will it take? • What must happen next on this job – subsequent steps? • Crane layout plans and lift plans
2.3	Develop Scope of Work	All scopes of work are to be developed following updated planning reviews. The development considers: • Jobs are developed against acceptance criteria defined in the approved options analysis. • Verify the availability of specialist equipment and tools, such as cranes. • Long lead items, including manufacturing of parts. • Business risk assessment is updated. • Critical path is verified for schedule conflict, group interactions or scope creep which could impact the duration. • On the completion of the final scope assessment, tasks are to be approved as shutdown activities.
2.4	Develop Budget	The first pass budget is to be developed. If Capital Works are to be included in the shutdown, then the funds need to be requested and approved as per the Glencore Copper Capital Management Framework.

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2.5	Confirm Long Lead Items	In collaboration with the supply department long lead items are to be identified and appropriate action taken to ensure the items are available on site prior to the planned task execution.
2.6	Develop Work Standards	Task and project owners are accountable to ensure the development, completion and submission of the following processes and documentation. • Detailed tasks with duration. • Resources required (supervisors, trade type, numbers required and equipment). • Parts and Materials needed. • Dependencies on other tasks for either commencement, consecutive progress, or completion. • Identification of potential hazards. • Permits and isolations required. • Any hold or inspection points required for Quality Assurance of work. • Specifications or Acceptable Standards the work must be completed to, and, • Applicable technical drawings or maintenance procedures. All documentation shall be completed and submitted before the mobilisation phase of the shutdown.
2.7	Complete Health, Safety, Environment and Community Plan	The purpose of this document is to define the Health, Safety, Environmental and Community (HSEC) Management Plan that will be in effect during the shutdown and all its associated activities. The HSEC management plan is to include: • HSEC organisation structure for the shutdown • Schedule of safety review meetings • Incident investigation procedures • Induction arrangements and training • Permitting procedures • Isolation procedures • Emergency response procedures • Hazard analysis identification, risk analysis and JSEA requirements.
2.8	Finalise Execution Philosophy	Following the shutdown scope approval process, the Shutdown Planner/Specialist is to finalise the execution philosophy by completing the following: • Level resources across the site shutdown and consider balance of internal vs external workforce. • Plan so that resources can be progressively released as workload drops off, after considering the opportunity to perform additional work. • Shutdown Managers' approval/sign off for medium risks. • General Manager approval/sign off for high and extreme risks. • Update shutdown master schedule and workbook • Any additional work must have a risk assessment conducted and approved by the relevant manager as per delegation of authority. Shutdown HSEC Plan including lesson learnt from previous shutdowns.
2.9	Constrain Schedule	Update the shutdown works package and master schedule. The master schedule is to be constrained in preparation for the final safety, interaction and shutdown risk assessment. Constraints include: • Total shutdown time • Resources

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		• Total budget actual against forecast
2.10	Conduct Readiness Review	The final phase prior to the shutdown optimisation is to ensure all tasks are ready to progress to the shutdown risk assessment. • Update readiness review checklist • Take go/no-go decision to move next step. • If tasks and/or stakeholders are not prepared, deficiencies will be escalated to the Shutdown Manager and relevant Manager for review.

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4.1.3 Shutdown Optimised – Level 3

Shutdown Optimised is the consolidation of the presented shutdown plans into a fully optimised, resourced and challenged process. (From plan to process) This phase is completed between the business risk assessment and the final shutdown risk assessment.

Key items in the optimisation phase are:

- Shutdown duration and critical path locked.
- Resources are confirmed and engaged.
- Materials and components are booked out or received.
- Risk management processes completed – interaction, safety and compliance.

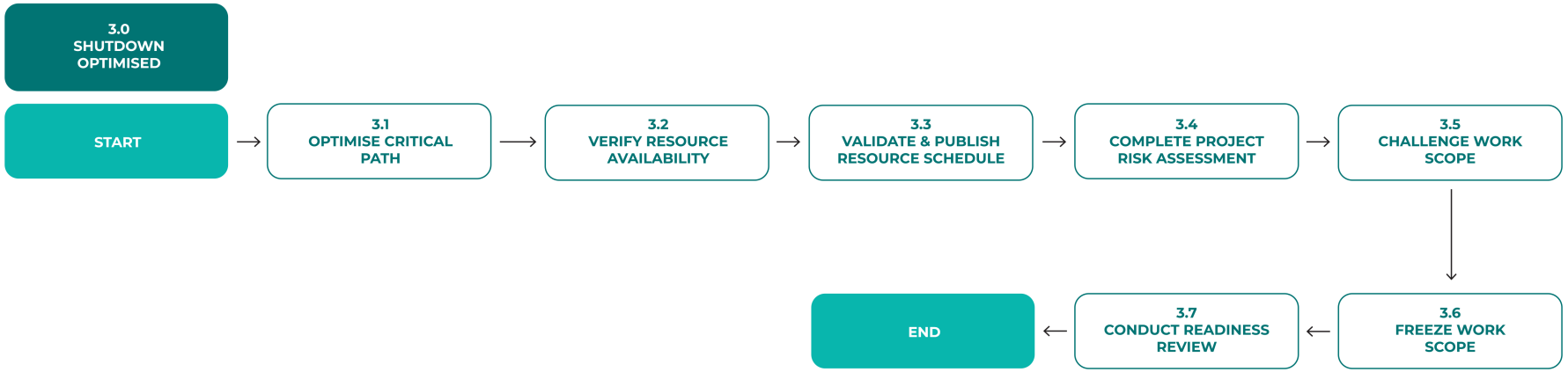


FIGURE 4-4 SHUTDOWN OPTIMISED PROCESS

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TABLE 4-3 SHUTDOWN OPTIMISED PROCESS DESCRIPTION

PROCESS STEP	PROCESS DESCRIPTION	DESCRIPTION
3.1	Optimise Critical Path	Where processes can be improved, aligned or started earlier, the schedule is to be updated. • Review each task to see if optimisation is possible. • Review the possibility of the equipment or component to be by-passed during normal operation (start early) • Review the possibility to realign task outside of the critical path (Parallel task) • Consider and review critical paths in series. • Once the schedule and critical path is optimised all tasks are to be validated for readiness, resource alignment, and preparation for risk review.
3.2	Verify Resource Availability	Shutdown teams together with the Shutdown Planner/Specialist shall align resources and resource utilisation and is site specific, suggested activities include, but not limited to: • Reduce peak resource requirements and smooth out period-to-period assignments using resource levelling. • Confirm nature of resource support – internal and external. • Where possible return contractors should be targeted, this will provide greater execution assurance and reduce the induction effort. • Rosters established. • Accommodation and support infrastructure requirements locked and booked. • Supervisors established and approved. • Site training schedule established. Individual task owners are to update their respective schedules and complete resource levelling. Schedule updates by task owners is to be provided to the Shutdown Planner/Specialist for inclusion into the final published shutdown schedule.
3.3	Complete Shutdown Risk Assessment	Key items captured in the risk review will include but not limited to: • Evaluate the timing of each task. • Look for potential interactions between work groups. • Identify safety risks. • Identify high risk work and control methods. • Define mitigation actions. On completion of the risk review process, the Shutdown Planner/Specialist will publish the Shutdown Risk Assessment.
3.4	Validate and Publish Fully Resourced Schedule	Collate all task specific information and update final Shutdown schedule. • Enter scope of work in the Shutdown Gantt. • Link job durations in the correct logic to form the shutdown Gantt chart. • Define the critical path and shutdown duration. • Optimise the critical path to minimise the shutdown duration
3.5	Challenge Work Scope	The Shutdown Planner/Specialist is responsible to complete the following key items: • Challenge rational for inclusion into the shutdown. • Assess resource availability.

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		• Assess critical path impact. • Obtain escalation approval where required. • Assess job against acceptance criteria. • Check constraints are not exceeded. • Changes to the shutdown duration to be approved. • Remove line items from schedule. • All tasks added/updated; execution plan updated. Any task that requires access to a shutdown will be identified using Maintenance Work Management and Capital Management Framework processes. The initial request is to be rejected if received after Shutdown Risk Assessment finalisation. Tasks then need to be reviewed with the task owner. Should the task be further rejected based on unresolvable impacts, the issue is to be escalate to the shutdown manager for review. General Manager sign off is required for emergent task inclusion. See Appendix A for example of Emergent Work form.
3.6	Freeze Work Scope	No additional tasks are permitted for inclusion in the shutdown. The Shutdown Planner/Specialist to complete the following: • Complete and issue shutdown execution plan for sign off. • Communicate shutdown overview to stakeholders. • If the estimated critical path is greater than the budgeted shutdown duration this needs to be escalated for approval to the relevant manager, due to the Production Plan impact. Tasks later identified as emergent works post shutdown risk assessment will require authorisation under current emergent work and Delegation of Authority processes.
3.7	Conduct Readiness Review	Review all processes and documentation supporting the planned shutdown. Where individual tasks are incomplete for this phase of the shutdown management process, an escalation will be raised with the task owner and respective manager. Key outcomes for this phase are: • Update the readiness review checklist. • Take go/no-go decision to move to the next step. • Incomplete tasks may be removed from the shutdown planned activities.

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4.1.4 Shutdown Mobilised – Level 3

Mobilise shutdown is the final phase of the shutdown management process before the Shutdown commences.

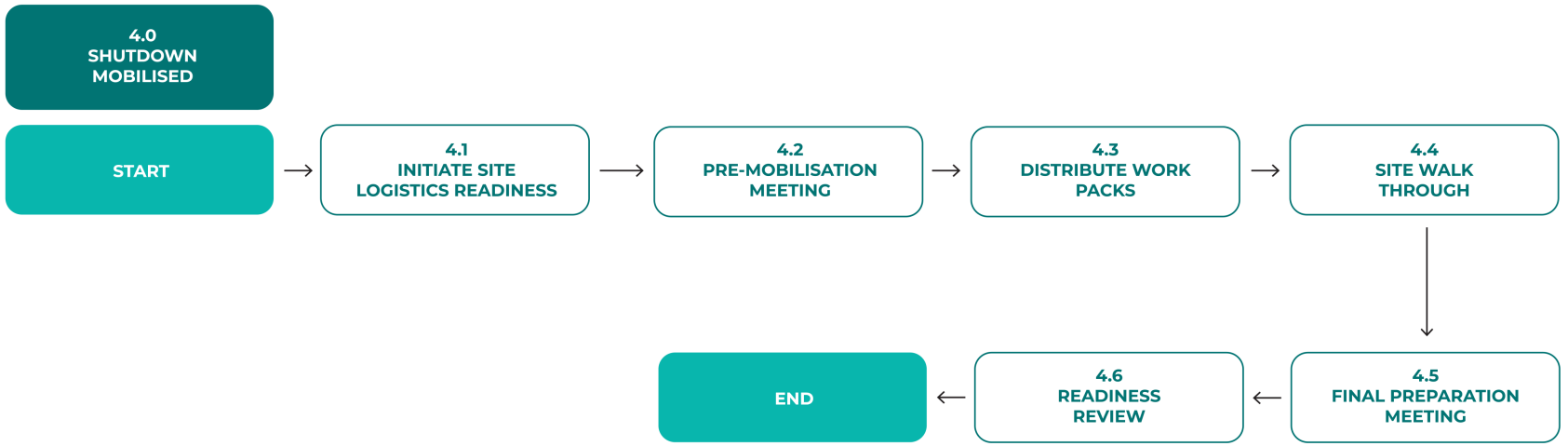


FIGURE 4-5 SHUTDOWN MOBILISED PROCESS

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TABLE 4-4 SHUTDOWN MOBILISED PROCESS DESCRIPTION

PROCESS STEP	PROCESS DESCRIPTION	DESCRIPTION
4.1	Initiate Site Logistics Readiness	Ensure the following requirements are met as part of the site readiness: • Site laydown map is completed. • Suitability of shutdown pad/site • Temporary facilities set-up, such as workspace, accommodation, messing, cribbing etc. • Crane lift plans are prepared and submitted. • Traffic management plans are submitted for review. • Operational requirements are aligned to the shutdown (machine interface, existing operational requirements and constraints are considered and understood) Other items for consideration could include: • Power • Lighting • Piping Develop Contractor onboarding plan and confirm contractor skills and certifications i.e., trade papers, confined space, working at heights etc. Any amendments or revisions required to the above documents/plans are to be addressed at the Mobilisation meeting.
4.2	Conduct Pre-Mobilisation Meetings	Separate meetings for internal and external (contractor) key stakeholder shall take place. Ensure the following pre-requisites are available for review by the key stakeholder group. Pre-requisites to be presented at the Mobilisation meeting are: • Laydown Plans • Lift Plans • Safe Work Methods Statements and Risk Assessments • Traffic Management Plans (where required) The following activities to be addressed: • Present the final approved shutdown schedule. Review meeting pre-requisite documentation (see above) • Confirm Supervisors and key personnel contact details for the shutdown. • Review/confirm isolation plan. • Review and update action plan from the Shutdown Risk Assessment • Finalise shutdown documentation. Issue the isolation plan to the authorised isolators for completion of isolation permits, lock boxes and isolation processes. • Finalise contractor on-boarding plan.
4.3	Complete and Distribute Work Packs	Asset owners, execution teams and respective managers are responsible to complete work packs for their respective tasks. Work packs shall contain information relevant to each task being conducted and as a minimum consider the following: • Title page with job description and CMMS work order number • Detailed description of the work • Work schedule

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		• Safety documentation – risk assessment, Safe Work Method Statement (SWMS) and procedures • Inspection test plan, Quality Control (QC) sheets. • Permit requirements. • Drawings, layout plans, Lift plans • Toolbox meeting sheets • Safety Data Sheets (SDS) • Job completion form The overarching shutdown work package is completed by the Shutdown Specialist/Planner for inclusion in the Shutdown Safety Management Plan (SMP). The SMP is completed for sign off 1 week prior to the commencement of the shutdown. On completion of sign off, the Shutdown Specialist/Planner will retain the works package until the day of commencement, when they are distributed.
4.4	Conduct Site Walk Through	Conduct a pre-walk-through meeting with the Shutdown and Safety Supervisors to confirm HSEC Plan and Scope of Work. Carry out a site walk through with key stakeholders. Review the site layout, potential clashes and consult the Shutdown Risk Assessment. Communicate the requirements of the shutdown and ensure personnel sign onto the safety management plan. Shutdown packs are handed to shutdown supervisors.
4.5	Conduct Final Preparation Meeting	Shutdown and Operational key stakeholders to meet for final meeting prior to Execution phase.
4.6	Conduct Readiness Review	Review the readiness checklist with Shutdown and Operational teams to ensure all items are available and approved before moving into execution. Items to be reviewed as part of the readiness checklist include: • Isolation plan, owners of isolations • Layout plan • Operational requirement and constraints are aligned. • Site preparedness, ground conditions • Supervisor details • Resource availability check • Review risk assessment action items and ensure completion. • Final combined stakeholder engagement prior to work execution. • Tracking of long lead items • Spares, parts and consumables available

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4.1.5 Shutdown Executed and Controlled

The purpose of the shutdown work execution and control phase is to provide a process to control the shutdown from operations handover, through shutdown execution and finally returning the plant and equipment to service.

The shutdown work execution and control phase establishes the standards of behaviour required to deliver the shutdown objectives. It is specifically concerned with the mobilisation and management of resources, the monitoring of activities to ensure they achieve standards required in a safe and proper manner. Teams must communicate, resolve issues, and provide updates among themselves and the management team on a daily basis. The visual representation of the performance and schedule allows close tracking of the shutdown progress. This will help to ensure the information needed for effective problem solving, issue escalation and decision making is available, where required.

4.1.5.1 Performance Requirement

The execution phase of tasks at Glencore Copper shall comply with the requirements set out in Element 08 Maintenance Work Management.

In addition to the shutdown work execution and control phase for each task, the following key points shall be actioned:

- All task owners shall carry out risk control measures prior to commencing their respective tasks or projects. Actions are identified in the shutdown risk assessment.
- Task owner provides updates on a shift basis. Updates are expressed as a percentage complete and are reported to the Shutdown Planner/Specialist
- The task owner shall inform the Shutdown Planner/Specialist of all incidents during the shutdown. Incident control shall comply with the Glencore Cu Incident Management Procedure.
- Task owner shall maintain schedule compliance during the shutdown. Where the schedule is altered or scope change occurs the task owner shall follow the Emergent Work Process, develop a Scope of Work and Budget for the work and seek approval to proceed through the Shutdown Planner/Specialist and escalated to the respective manager as per Delegation of Authority.
- The task owner shall inform the Shutdown Planner/Specialist where budget changes are identified.
- The task owner shall ensure the Shutdown Planner/Specialists is informed when the task is completed. In addition, the task owner shall ensure all preparation, handover and quality assurance is completed prior to the commissioning phase.
- The asset owner shall ensure the plant is commissioned to functional specification requirements.
- The Shutdown Planner/Specialist shall ensure all plant is checked, including critical equipment that did not form part of the Shutdown, for commissioning readiness and complete plant handover to the commissioning team.
- On completion of commissioning the Shutdown Planner/Specialist will be informed and carry out operations handover.

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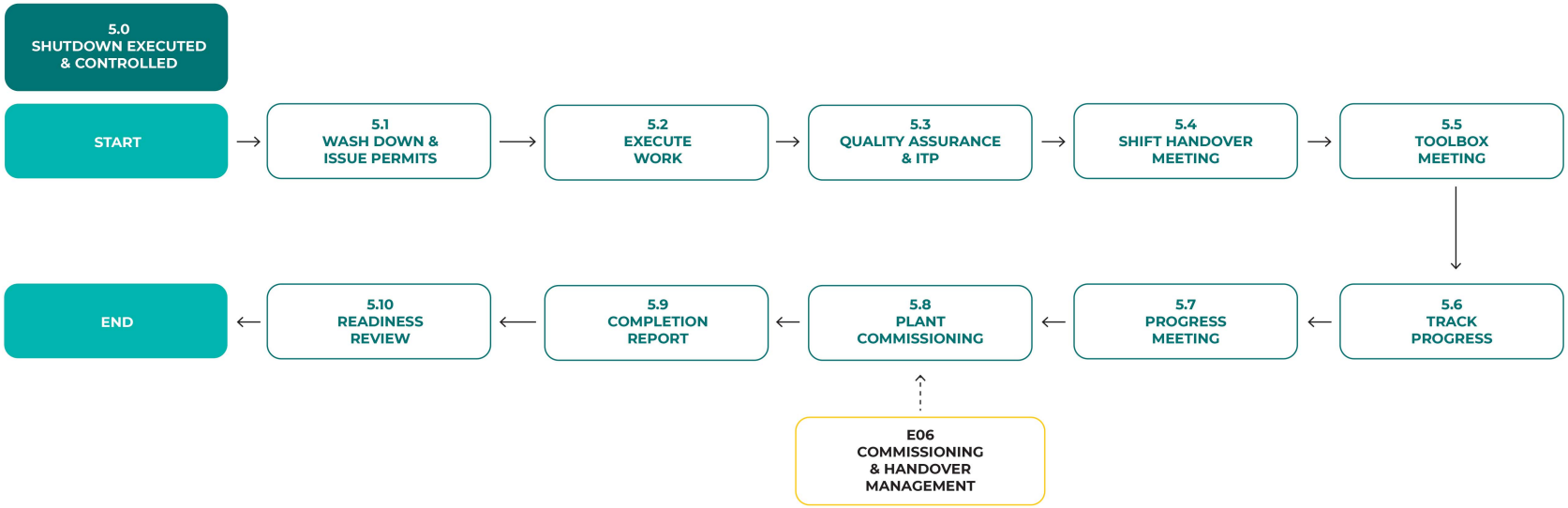


FIGURE 4-6 SHUTDOWN EXECUTED AND CONTROLLED PROCESS

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TABLE 4-5 SHUTDOWN EXECUTED AND CONTROLLED PROCESS DESCRIPTION

PROCESS STEP	PROCESS DESCRIPTION	DESCRIPTION
5.1	Execute Washdown and Issue Permits	Each Stakeholder is to identify washdown requirements following the Shutdown Risk Assessment. Requirements are to be documented and issued to the relevant Operations person 2 weeks prior to the commencement of the shutdown. Isolations and work specific permits are to be completed post Shutdown Risk Assessment. • Isolations: > Complete plan > Authorised isolator receives > Isolation permit completed. • Permits: > Work permit completed > Permit reviewed > Permits activated • Control permits – Hot works – Working at heights – High Voltage (HV) – Grid mesh remove etc.
5.2	Execute Work	Shutdown work is to be executed as per the final approved schedule and procedure. Risks are to be assessed prior to task execution and actions taken if the post execution identified risks differ from actual assessed risk. Any deviation from the approved schedule must be reported to the Shutdown Planner/Specialist. Ensure work execution is in compliance with the HSEC Plan and critical controls are verified.
5.3	Quality Assurance and Inspection and Test Plan (ITP)	Individual teams are responsible to produce the following: • Detailed schedule outlining QA / QC points • Inspection test plans for tasks as defined in the individual shutdown requirements and for critical equipment that has been idle for the duration of the shutdown and did not form part of the Scope of Work. • Commissioning plans • Where hold points are identified in tasks, the stakeholder will complete and provide documented evidence of the activity. • Quality assurance documentation will be provided in the shutdown planning phase.
5.4	Conduct Shift Handover Meeting	A shift handover meeting is to be conducted in each work area between Shutdown and Operational supervisors to discuss the following items: • Safety share • Completed and planned work • Safety concerns for the shift • Change in isolation status. • Labour/Personnel • Non-compliance and reason • Emergent work • Interactions and shared service (cranes etc.) The meeting will be documented in the shutdown Shift Handover meeting form. Each participant is required to sign the attendance portion of the handover meeting minutes.
5.5	Conduct Toolbox Meeting	A toolbox meeting with the shutdown team is to be conducted by the area shutdown supervisor prior to each shift to discuss: • Safety share • Work completed in the last shift.

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		• Work for this shift, for example, safety issue, job progress from previous shift and work scope for this shift including confirming job scope and steps, risk controls etc. • Issues or concerns Each participant is required to sign the attendance portion of the toolbox meeting minutes.
5.6	Track Progress	• Track progress for each activity against the schedule on a daily basis. Progress is expressed as a percentage complete or duration remaining to complete an activity. Progress information is then included in a daily report and distributed to key stakeholders. It should also include actual cost against planned cost and any safety related issues in the period. • As jobs progress the critical path and near critical path are to be managed to ensure no slippage in the shutdown occurs. If deviation occurs or is predicted to occur actions plans are to be put in place to ensure activities are brought back to original timeline. • Late and emergent work requests are to be evaluated and those that meet the emergent work criteria are to be forwarded to the shutdown manager for approval, inclusive of the cost and risk associated with the emergent work. • Evaluate and respond to safety related issues as they occur.
5.7	Conduct Shutdown Progress Meeting	A shutdown progress meeting is to be conducted daily to include the following items: • Planned v actual work and budget. • Review high risk activities, SWMS and relevant risk assessments. • Mobilisation of equipment • Shutdown update • Safety and environmental issues and general business • Progress on Gantt chart, milestones, S-curve, review estimate at completion (EAC) • Variation register • Upcoming possible time impact events (blasting) • One day and three day look ahead as applicable • Outstanding parts and equipment • Progress of actions plans. • Review progress issues. • Escalate non-conformances.
5.8	Execute Plant Commissioning	Refer to Glencore Copper Asset Management Framework E06 Commissioning and Handover Management Manual. Commissioning of the plant and/or equipment is to be completed by the individual teams specific to their respective scope of work. On completion of task commissioning, housekeeping and handover readiness review, the plant and/or equipment is handed over to the asset owner for approval. The asset owner on acceptance of the plant and/or equipment will complete commissioning processes as per the commissioning plan, together with required shutdown execution person/s.
5.9	Prepare Completion Report	A collated list of activities performed during execution is sent to Operational stakeholders on completion of the shutdown. This report will outline specific information relevant to each task or project that has been completed during the shutdown. The aim is to ensure maintenance personnel can fault find and understand key plant performance.

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		Incomplete scopes or punch list items will be identified and communicated to the asset owner. The handover of plant to operations will ensure components, equipment and systems are available to operate to the functional specifications, designed operating standards (OEM) and compliance standards. Should the equipment have restrictions or limitations, the asset owner is responsible to ensure the restrictions or limitations are communicated to Operations and include the following: • Clear procedure to operate. • Procedure accompanied by a risk assessment where required. • If changes to operational parameters exist, Operators will become familiar with the requirements. • Acceptance from Operations Manager.
5.10	Conduct Readiness Review	Review operational constraints to live plant and/or equipment where specific demobilisation activities interact. Where demobilisation activities limit equipment operability, appropriate control measure are to be identified and actions requested of task owners. i.e. • Long travel blocks for crane demobilisation. • Traffic management plan to cover machine movements restrictions and barricading. • Operations have received communications for any plant and/or equipment restrictions, access restrictions or potential delays on access routes. • Demobilisation plan is updated and communicated by the project or task owner, added to site noticeboard and issued to the Shutdown Planner/Specialist for alignment to the traffic management plan. • Housekeeping is in order post demobilisation.

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4.1.6 Shutdown Closed Out

Shutdown close out ensures shutdowns are completed and closed effectively. Completion of a shutdown does not happen at handover to operations or plant and/or equipment return to service. Shutdown completion ends after the following items are critically reviewed.

- Personnel safety, incidents and safety improvements.
- Task completion and carry over planning.
- Risk mitigation and task objectives.
- Punch list items.
- Budget planned against actual spend.
- Handover timing.
- Change management completed.
- CMMS updated and work orders closed.
- Plant reliability.
- Lessons learned.
- Shutdown improvements.

Shutdown closeout identifies the activities in the shutdown management process that starts at operations handover through to the shutdown closeout. At handover to operations, task demobilisation is planned and executed as a separate function to the shutdown. Closeout processes identify:

- Interactions with operational plant during demobilisation.
- Plant reliability post-handover (28 days)
- Post investment review process
- Carry over tasks that may require additional shutdowns.

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FIGURE 4-7 SHUTDOWN CLOSED OUT PROCESS

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TABLE 4-6 SHUTDOWN CLOSED OUT PROCESS DESCRIPTION

PROCESS STEP	PROCESS DESCRIPTION	DESCRIPTION
6.1	Execute Demobilisation Plan	Consolidate all task demobilisation plans into a central plan. The demobilisation plan is updated by the Shutdown Planner/Specialist based on individual task conformance against schedule. The demobilisation plan is activated prior to the shutdown handover to ensure readiness by work teams. Key items are considered: • Live plant commissioning activities and impacts • Restrictions to plant while demobilising (long travel limits or blocks) • Interruptions to site access routes • Ensuring the traffic management plans are in place and relevant. • The Shutdown Planner/Specialist together with the key stakeholders activates the demobilisation plan and completes the shutdown handover check list. • All unused spares and equipment are returned to the store in a timely manner. • Housekeeping is to be conducted to the required standard, waste and material are to be disposed of and stored accordingly.
6.2	Complete Closeout Meeting/s	There will likely be a number of closeout meetings with both internal and external stakeholders, including an area meeting, lessons learnt meeting and complete shutdown team meeting. Meetings shall be scaled according to the size and complexity of the shutdown, which should be assessed by each region/site. Internal meetings are to be conducted with key shutdown stakeholder and an external meeting with key contractor stakeholders to discuss lessons learnt and review performance. Typically, lessons learnt discussion will focus on: • Health, Safety and Environmental performance for each work group/area • Actual cost of the shutdown versus the planned cost • Actual duration of the shutdown versus the planned. • Emergent work completed and not completed. • Contractor performance • Materials and services management plan • Effectiveness of the ITP and QA plans inclusive of Work Packs. • Effectiveness of the people resource management plan. • Effectiveness of the shutdown organisation and role descriptions. • What was done well and not so well. Lesson learnt should be captured at the earliest opportunity, documented and action plans developed to ensure future shutdown improved performance. Factors external to the organisation are evaluate against • the business • technology and tools, • organisation • process and the people. KPI's that are derived from this Manual and site-specific measures are reviewed and scored.

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6.3	Prepare Shutdown Closeout Report	The following information is documented as part of shutdown closeout report: **Shutdown Completion** – Under this section of the report shutdown objectives, deliverables, change management implementation and outstanding items are tabled with each item evaluated for acceptance. Key stakeholders involved in the post implementation inspection are also listed to accompany the evaluation for acceptance. **Financial Summary** – Under this section of the report a cost summary is outlined tabling the shutdown budget, actual spend against planned expenditure and the final total forecast. Cost commentary will also form part of this section. **Lessons Learnt** – as discussed in Section 6.2 Any specific lessons learnt that can improve shutdowns, processes and/or scopes are evaluated as part of the continuous improvement process. **Customer Acceptance** - When approved this confirms acceptance by the customer and other relevant stakeholders. **Shutdown Closure Meeting** – Where a shutdown close-out meeting has occurred the shutdown closeout report will outline when the meeting was held and who was in attendance. **Shutdown Closure Meeting Minutes** – Where a close out meeting has been completed minutes will be attached as an appendix.
6.4	Finalise Shutdown Data and Information	All data and information shall be entered/updated in the relevant business system by the authorised responsible person/s. Repetitive jobs shall be updated by the responsible person and the CMMS updated.
6.5	Conduct Readiness Review	Collate incomplete scope items, punch lists and carry over tasks. The tasks will then be assigned to the relevant stakeholder. The task owner shall provide an updated schedule that identifies the new shutdown that the tasks are to be completed in. The Shutdown Planner/Specialist will then engage the long-term planner to assess the inclusions into the CMMS. Repetitive job scopes are to be updated as required in a timely manner so that improvement opportunities are captured. Determine timeline for next shutdown cycle to begin.

4.2 Process RACI Description

The RACI matrix below in Table 4-7 addresses the high-level functional elements of the Asset Management Leadership process. For the overall Asset Management RACI matrix, refer to the Asset Management Intranet page.

The RACI provides and indicative overview of the functional requirements for a shutdown, region/area/site will have their own specific requirements based on the size and complexity of the shutdown. This shall be taken into consideration when forming the shutdown team/s and developing resource requirements.

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The RACI is applicable to both Glencore Copper personnel and Glencore Copper contractors. Should there be a requirement for people to perform a different role in the shutdown from their normal business as usual role, the RACI for the shutdown role is to be followed.

Note: The Commercial Department does not appear in this high-level RACI, although their involvement will be required from time to time dependent on the nature of the variance.

The codes used in the RACI analysis are defined as follows:

TABLE 4-7 PROCESS RACI

R	Responsible	The "doer" – the individual(s) who will perform an activity or directs others to perform the activity.
A	Accountable	"The buck stops here" –the individual who ultimately "owns" the activity – there should only be one person
C	Consulted	The individual(s) who needs to provide feedback and contribute to the activity
I	Inform	The individual(s) who needs to be informed after a decision or action is taken if relevant. (via 1-way communication)

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Function / Process Step	General Manger	Shutdown Manager	Shutdown Specialist/ Planner	Shutdown Coordinator/ Area Coordinator	Operations Manager/ Supervisor	Shutdown Supervisor	Shutdown Trades	Health, Safety, Environment and Training	Cost Control Coordinator	Engineering Manager	Procurement and Supply Coordinator
1.0 Establish Shutdown											
1.1 Life of Asset Plan	A	R		C	C				C	C	
1.2 Establish Shutdown Organization	A	R		C	C				C	C	C
1.3 Develop Work Breakdown Structure		A	R	C	C	I		I	C	C	
1.4 Create Budget Requirements		A	R	C	C	I		I	C	C	C
1.5 Determine Shutdown Preparation Milestones		A	R	C	C	I		I	C	C	
1.6 Conduct Initial Shutdown Meeting		A	R	R	R	R		R	R	R	C
1.7 Conduct Shutdown Business Risk Assessment		A	R	R	R	R		R	R	R	C
1.8 Complete Readiness Review	A	R	R	R	R	C		C	C	C	C
2.0 Plan Shutdown											
2.1 Develop Scope of Work		A	R	R	R	C	C	C	C	C	
2.11 Are External Services Required?		A	C	C	C	C		I	C	C	
2.2 Obtain Further Planning Information		A	R	C	C	C	C	C	C	C	
2.3 Develop Scope of Work	A	R	R	R	R	C	C	C	C	C	
2.4 Develop Budget	C	A	R	R	R				R	R	C
2.5 Confirm Long Lead Items		A	R	I	I	I			R	I	R
2.6 Develop Work Standards		A	R	R	R	R	I	C	R	R	C
2.7 Complete Health, Safety, Environment and Community Plan		A	R	C	C	C	I	C	C	C	
2.8 Finalise Execution Philosophy	A	R	R	R	R	C	I	C	C	C	

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Function / Process Step	General Manger	Shutdown Manager	Shutdown Specialist/ Planner	Shutdown Coordinator/ Area Coordinator	Operations Manager/ Supervisor	Shutdown Supervisor	Shutdown Trades	Health, Safety , Environment and Training	Cost Control Coordinator	Engineering Manager	Procurement and Supply Coordinator
2.5 Constrain Schedule		A	R	C	C	I	I	C	C	C	C
2.6 Conduct Readiness Review	C	A	R	C	C	C	I	C	C	C	C
3.0 Optimise Shutdown											
3.1 Optimise Critical Path		A	R	R	C	C	I	C	C	C	C
3.2 Verify Resource Availability		A	R	R	R	C	I	C	R	R	I
3.3 Complete Shutdown Risk Assessment	C	A	R	R	R	C	I	R	R	R	C
3.4 Validate and Publish Resourced Schedule	I	A	R	C	C	I	I	I	I	I	
3.5 Challenge Work Scope		A	R	C	C	C		I	C	C	C
3.6 Freeze Work Scope	I	A	R	C	C	C		I	C	C	I
3.7 Conduct Readiness Review	I	A	R	C	C	C	I	C	C	C	C
4.1 Initiate Site Logistics Readiness	C	A	R	C	C	C		C	R	R	C
4.2 Conduct Pre-Mobilisation Meetings	C	A	R	C	C	I		C	C	C	C
4.3 Complete and Distribute Work Packs		A	R	R	C	R	C	C	R	R	
4.4 Site Walk Through	C	A	R	C	C	R	I	R	R	R	
4.5 Conduct Final Preparation Meeting	C	A	R	R	R	C	I	R	R	R	R
4.6 Conduct Readiness Review	I	A	R	C	C	C	I	C	C	C	C
5.1 Washdowns and Issue Permits		A	R	R	R	R	I	C	C	C	
5.2 Execute Work	C	A	R	R	R	R	R	R	R	R	R
5.3 Quality Assurance and ITP	C	A	R	R	R	R	I	C	R	R	C
5.4 Shift Handover and Prestart Meeting		A	C	C	R	R	C	C	R	R	C
5.5 Conduct Toolbox Meeting		A	C	C	R	R	C	C	R	R	C
5.6 Track Progress	C	A	R	R	R	C	C	C	R	R	C

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Shutdowns & Overhauls Management

Function / Process Step	General Manger	Shutdown Manager	Shutdown Specialist/ Planner	Shutdown Coordinator/ Area Coordinator	Operations Manager/ Supervisor	Shutdown Supervisor	Shutdown Trades	Health, Safety , Environment and Training	Cost Control Coordinator	Engineering Manager	Procurement and Supply Coordinator
5.7 Conduct Shutdown Progress Meeting	C	A	R	R	R	C	I	R	R	R	R
5.8 Plant Commissioning	C	A	C	R	R	R	R	R	R	R	R
5.9 Completion Report	C	A	R	R	R	R	C	R	R	R	R
5.10 Conduct Readiness review	C	A	R	R	R	R	C	R	R	R	R
6.1 Execute Demobilisation Plan		A	R	R	R	R	R	R	R	R	R
6.2 Closeout Meeting	C	A	R	R	R	R	C	R	R	R	R
6.3 Closeout Report		A	R	C	C	C	C	R	R	R	C
6.4 Shutdown Mater Data		A	R	C	C	C	I	C	C	C	C
6.5 Conduct Readiness Review	C	A	R	R	R	C	I	C	R	R	R

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TABLE 4-8 FUNCTIONAL DESCRIPTIONS

Function	Description
General Manager	• Aggregate and integrate Asset Management Framework with accountability for its implementation across the site. • Improve organizational culture through effective engagement and motivation of people across the site. • Develop and deliver Life of Asset Plan and Budgets • Sponsor Site Strategic Asset Management Plan • Control of maturity evolution across the site. • Ensures the Shutdown and Overhaul process is embedded, monitored and improved across the area/site.
Shutdown Manager	• Ensure a robust Shutdown Plan exists, is aligned to business needs and is delivered. • Provide feedback to the Steering Committee on progress or status as required. • Ensures all tasks required in the shutdown are raised and managed. • Manage the Change Approval Process for new and emergent work. • Identify Shutdown resources requirements to ensure the Plan is executed safely and effectively. • Accountable for the Shutdown Budget and schedule • Assures multi-departmental involvement and cross-functional communication. • Manages Risks by ensuring the identification and mitigations systems are in place. • Ensure continual improvement and development of the Shutdown Process. • Shutdown Compliance milestones and tollgates. • Final Shutdown report is developed and actioned. • Contractors are identified and engaged. • Mobilisation and demobilisation plans developed and executed. • Chairs the pre-shutdown meeting/s. • Chairs daily shutdown meetings. • Chairs the close out shutdown meeting/s.
Shutdown Planner/Specialist	• Consolidating and reporting on planning worklist • Owner of shutdown Gantt development and updating. • Consolidating of job plans into the Gantt. • Identifies and tracks long lead items. • Overall Planning Progress reporting • Develops floor plans for use by planners and then managing in Gantt. • Sets up and leads meetings for Critical and near-critical path job evaluation and optimisation. • Identifies and mitigates conflicts that are present in the Shutdown Gantt. • Organises and Chairs review meetings required for Gantt development and refinement. • Updates Shutdown data and information as directed. • Updates repetitive tasks in CMMS.

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Shutdown Coordinator or Shutdown Area Coordinator	• Responsible for Content and Quality of Worklists and Job Plans. • Work with the Shutdown Planner Specialist on a regular basis to ensure Worklist and Job Plans are properly compiled and to ensure details are properly entered in the Master Gantt with all the correct relationships and timing. Provide Schedule oversight to manage key interfaces between contractors, jobs and work in their area. • Preliminary Critical Path are identified and managed. • High risk jobs from safety perspective are prepared and managed. • Ensure shutdown tasks are executed safely, on time and budget within their area of responsibility. • Escalates emergent work requirements to the respective manager of their area of responsibility. • Attend required shutdown meetings. • Provides shift shutdown updates to Shutdown Planner/Specialist. • Ensures plant and equipment is handed back to operations in good clean condition ready for commissioning. • Deployed as part of the commissioning team. • Update's shutdown data and information as directed. • Provides input for shutdown report.
Shutdown Supervisor	• Ensure input from their department into worklists and the shutdown plan. • Ensure work orders are raised for equipment repairs and maintenance activities to enable reliable operations to the next scheduled down period. • Challenge work being raised from the maintenance department is required and aligns with the shutdown objectives and strategy. • Supply resources to the shutdown planning team and manage operations to allow release from normal duties for shutdown planning, training, required prework and execution. • Supervises and controls the shutdown activities, ensure zero incidents and work quality and ensuring housekeeping is in order before, during and after the shutdown. • Support the shutdown management process. • Attend required shutdown planning meetings. • Supervises the execution of their responsible shutdown tasks, ensuring each task is completed within planning constraints. • Conducts prestart toolbox meetings with team under their supervision. • Ensures all safety protocols are adhered to and the required PPE is available and worn. • Escalates emergent work to Shutdown Coordinator or Shutdown Area Coordinator. • Provides shift updates to the Shutdown Coordinator or Shutdown Area Coordinator.
Operations Manager/Supervisor	• Development and approval of the ramp down production plan including process shutdown activities and pre-shutdown plant positioning. • Development and approval of the ramp up production plan including prestart production activities, commissioning and test considerations and the process and ramp up production steps. • Ensure work orders are raised for equipment cleaning and shutdown operational activities to enable reliable operations to the next scheduled down period. • Ensure plant and equipment availability as per the shutdown plan. • Development and customisation of the area handover checklists of handover at the beginning and end of the execution phase.

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	• Coordinating the involvement of commissioning and process engineering requirements in the ramp down and ramp up plans. • Ensure all permits are issued, isolation plan in place and managed. • Attend required shutdown planning meetings.
Engineering Manager	• Development and approval of engineering activities for the shutdown and ensure their integration into the shutdown plan. • Engineering tasks are completed to schedule. • Ensure work orders are raised for engineering inspection activities to enable reliable operations to the next scheduled down period. • Development and customization of any commissioning plans and ensuring the requirements are built into the ramp down and ramp up plans. • Attend required shutdown planning meetings. • Ensures Capital Projects are completed safely, on time and budget. • Plant and equipment in handed back to operation in good clean condition. • Forms part of the commissioning team as required.
Cost Control Coordinator	• Ensures emergent work is fully costed, planned and authorised by the relevant manager. • Ensures contractual arrangements are in place for emergent work. • Ensures spares and equipment are available for emergent work. • Updates budget planned versus actual information. • Liaise with Procurement and Supply Chain coordinator to procure and expedite emergent work parts, materials and contractors as required.
Health, Safety, Environment and Training	• Build Health, Safety and Environmental Management plan. • Participates in Safety Risk reviews. • Audits Shutdown activities, ensuring they are executed in a safe and controlled manner. • Induction management plan developed and executed. • Contractor credential identified and verified. • Training needs identified and delivered. • Site access assessed and granted.
Procurement and Supply Chain Coordinator	• Ensures parts and materials are available as per the shutdown plan. • Liaise with Cost Control Coordinator to procure and expedite emergent work parts, materials and contractors as required. • Ensure any returned parts and materials are logged back into the CMMS and allocated back to the store.
Shutdown Trades	• Executes shutdown tasks as directed in a safe and timely manner. • Attends shutdown meetings as directed.

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5. Deployment

The deployment of the Shutdown and Overhaul Management approach across all business areas is through:

- Awareness presentation
- Position description alignment
- Competency assessment
- Training, coaching, and mentoring

5.1 Awareness Presentation

This Manual is supported by a slide presentation and video to help facilitate communication and to raise awareness on how it contributes to the Asset Management Framework and business success.

5.2 Position Description Alignment

To effectively deploy this Manual, the essential process steps and activities need to be diligently mapped and assigned to approved positions within Glencore Copper.

The RACI matrix assists in ensuring everyone is working towards the same goal. It clarifies, not only the individual's responsibilities, but also that of others. It has proven to be an effective tool to quickly understand and close the gaps between current and desired practices and business process activities.

Position descriptions including KPIs need to be aligned to the RACI matrix and measures of success. This is the basis for ongoing performance reviews and the creation and execution of individual development plans.

5.3 Competency Assessment

This Manual is supported by an Asset Management competencies assessment process and each site is responsible for assessing Asset Management competencies which is used to:

- Guide recruitment, selection, development, and deployment of people in asset management roles across the company
- Provide employees responsible for managing assets with feedback, development, coaching and recognition for their performance.
- Map, build and manage the asset management talent pool across the company.
- Identify strengths and development needs in asset management practitioners.

5.4 Training, Coaching and Mentoring

Coaching and mentoring is another way to assist employees improve their understanding of their role and to improve performance.

Coaching is generally about improving performance in the short term where mentoring is about achieving high performance and career aspirations over a longer term.

Coaching or mentoring is required when a person is not currently working competently in their role. The build-up of skills to perform efficiently and effectively requires a large investment of time and cooperation.

Coaches and Mentors need to be carefully selected and must be committed for the long term to effect maximum success.

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6. Measures of Success

Refer to Glencore Copper's AM Manual E16 Reporting and Performance Management Manual for current KPI's

TABLE 6-1 MEASURES OF SUCCESS

Metric	Jobs Cancelled (Number)
Definition	Jobs that are cancelled after the shutdown schedule has been closed or during the shutdown
Performance Indicator	No jobs are to be cancelled
Calculation	
Rationale	This KPI provides a measure of the performance of the Shutdown Planning process. If jobs have to be cancelled, then this is a reflection that the planning was not commenced at sufficient time out from the shutdown to allow for all activities to be executed or all resources to be available. Resources for the shutdown include people, spares, materials, cranes, scaffolding, tool, lifting equipment, PPE, etc.

Metric	Emergent Work (%)
Definition	% of Emergent Work Manhours against the total of all planned shutdown manhours
Performance Indicator	<5% Emergent Work Hours
Calculation	(Emergent Work Hours/Total Planned Shutdown Work Hours)/100
Rationale	Significant Emergent work is an indicator that all risks for the shutdown have not been assessed thoroughly and that no contingencies have been put in place to reduce the risk to tolerable levels. Significant Emergent work will cause scope creep (time and cost) and hence potential start up delay.

Metric	Start Up Compliance (%)
Definition	% of time greater than the planned shutdown time from full capacity (pre-shut) to full capacity (post shut)
Performance Indicator	On time as per the shutdown plan
Calculation	(Total time over planned time/Total Planned Time)/100
Rationale	This KPI provides a means by which Shutdown Planning can be measured. The shutdown consists of 3 main phases Ramp Down, Execution and Ramp up (full capacity), and all 3-phases are to be included in the shutdown planning process. This promotes a collaborative and one ownership approach between functions to ensure that the start-up and full capacity occurs as per the shutdown plan.

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Metric	Shutdown Schedule Compliance (%)
Definition	% of scheduled shutdown work orders compared to the total shutdown work orders completed
Performance Indicator	100% Compliance to Schedule
Calculation	[Number of scheduled work orders completed in the shutdown / Total number of work orders planned in the shutdown] x 100
Rationale	This is important as it is used to monitor the effectiveness of the shutdown planning and scheduling process and the completion of work for the shutdown.

Metric	Shutdown Budget Compliance (%)
Definition	Actual Budget vs Planned Budget
Performance Indicator	Actual Budget +/- 5% of the Planned Budget
Calculation	(Actual Budget/Planned Budget)/100
Rationale	This is important as it is used to monitor the effectiveness of the shutdown planning and scheduling process and the compliance to the planned budget for the shutdown.

Metric	Post Shutdown Availability (%)
Definition	Plant and equipment availability 28 days post shutdown
Performance Indicator	Availability is at or better than forecast availability in the period.
Calculation	
Rationale	Rework can often become an issue post shutdown, so measuring plant and equipment availability post shutdown Is a way in which shutdown quality performance can be measured. Forecast availability is the budgeted time the plant and equipment is to be available in the period, generally measured monthly. If availability drops below forecast this could provide an indication that the quality of the shutdown work has not been executed to the required standard.

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Metric	Shutdown Close Out Report Exists
Definition	A measure to see if a shutdown report has been generated post shut.
Performance Indicator	Yes/No
Calculation	
Rationale	A shutdown report is an excellent tool for capturing shutdown performance and shutdown lessons learnt and every shutdown is to have a post shutdown report developed.

Metric	Lessons Learnt Implemented
Definition	Lessons learnt from the shutdown are to be progressed to actions and an implementation plan developed and executed.
Performance Indicator	100% Lesson Learnt Implemented by the next shutdown
Calculation	
Rationale	Lessons learnt provides insight as to what went well and what did not go well during the course of the shutdown. This provides a valuable means by which success can be replicated or improvement opportunities can be identified and implemented.

Metric	Shutdown Work Packs Documentation
Definition	All shutdown work packs have the required documentation developed and attached to the work pack prior to the work pack being issued.
Performance Indicator	100% of tasks have work packs and all documentation is present
Calculation	
Rationale	Ensures that shutdowns are risk assessed, planned and executed in a safe and controlled manner.

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Metric	Shutdown Tasks with Signed Documentation
Definition	All shutdown tasks complete have the appropriate safety and environmental documentation signed and attached to the work order at job commencement.
Performance Indicator	100% of tasks have signed safety and environmental documentation (for example - JSEA, SLAM) attached.
Calculation	
Rationale	Each task has to be assessed before the task can commence where documents such as JSEA's, SLAM's, Take 5's and may have to be updated as the task progresses. These documents are to be signed and attached to the work order in order to satisfy the requirement that assessment regarding safety and the environment have been conducted prior to job commencement.

Metric	Shutdown Fatal Hazard Protocol (FHP) Tasks Observed
Definition	FHP tasks that have commenced are to be observed each shift to ensure that the task is executed in a safe and controlled manner and that all documentation is present and correct.
Performance Indicator	100% of FHP tasks that have commenced to be observed each shift.
Calculation	
Rationale	Tasks identified as being FHP pose the greatest risk to the shutdown execution team, therefore close scrutiny of these task is essential to protect against injury or harm.

7. Improvement

The process to review and continually improve the approach and deployment for Shutdown and Overhaul Management is scheduled Annually, or when there is a sudden change in the Operating Context. As the organization matures a longer review period could be considered.

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8. References

The following documents provide detailed support material that should be read in conjunction with this Standard to provide a comprehensive understanding of Shutdown and Overhaul Management.

TABLE 8-1 REFERENCES

DOCUMENT TITLE	DESCRIPTION
Asset Management Framework	This document defines the overall direction of asset management across Glencore Copper.
Asset Management Metrics Standard	This document defines the measures/KPIs, the method of calculation, the intent and behaviours that each measure is aimed at.
Asset Management – Shutdowns and Overhauls Management Guideline	This document defines Glencore Copper's approach to managing the shutdowns, major installations and overhauls, and major maintenance projects.

9. Revision History

VERSION NO.	REASON FOR REVISION	DATE
0.1	New Manual to support the Asset Management Framework	18/03/2021
0.2	Gather Documents and Create Draft Manual	27/03/2021
0.3	Review and Signoff 1st Draft Manual with Sites	03/04/2021
0.4	Reformat and Issue for Final Review	13/05/2021
1.0	Update Site Signoff and Publish	11/10/2021

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10. Appendix A – Emergent Work Form Example

Emergent Work Approval Form

Planned Shutdown

Sector _____ **Requested by :** _____

Equipement : _____ **Date requested :** _____

Work Order Number : _____ **Hours of Work :** _____

Job Description : _____

A job is considered out of time when it is requested after the job has been identified.

Scope of Work

Estimated Material Cost
$

Estimate Labour Cost
$

Cost Allocation
Cost Centre

Estimated Duration		
Start Date:	Finish Date:	Days/Hrs:

Offsite Repair	
Yes:	No:

	Is the work really required?	Yes	No
1.01	Is the work required to eliminate a health and safety risk in operation		
1.02	Is the work required to eliminate an environmental risk in operation		
1.03	Is the work required in order to avoid an unplanned production shutdown		
1.04	Is the work required in order to avoid a reduction in the rate of operation		
1.05	Is the work required in order to avoid additional costs		
1.06	Is work required now		
	Planning (6M)	**Yes**	**No**
2.01	Is the workforce expected and available		
2.02	Is the equipment or will be on the site 3 days before the shutdown		
2.03	Is the method where will be well defined 2 days before stopping		
2.04	Are the tools, machinery, services planned and available		
2.05	Are the quality controls (Measure) defined for the works		
2.06	Is the working environment known and is or will be controlled		
	Coordination	**Yes**	**No**
3.01	Does the work conflict with other work		
3.02	[1]Is the working time greater than half of the planned duration of the stoppage? (including padlocking, preheating and any work to be carried out before or after this		
3.03	Does the duration of work influence the duration of the stoppage		
3.04	Is the working time more than 10 hours		
3.05	Does the work require changing the coordination of other planned work		
	Approval	**Yes**	**No**
4.01	Does the area operations superintendent approve the execution of the work		
4.02	Does the area maintenance superintendent approve the execution of the work		
4.03	Does the outage management superintendent approve the execution of the work		

Good Practise
If one or more red boxes are checked, it is essential to properly analyze the impact of the work added on the entire shutdown.
[1] It is also necessary to ensure that the planning and preparation meet the standard of preparation for shutdown works
It is recommended to validate whether it would be more convenient to make another stop for this work
[2]Emergency or last minute work requires increased supervision, so this supervision must be provided.

Remarks

Signatures		
Operations Superintendent Date	Maintenance Superintendent Date	Shutdown Superintendent Date

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11. Appendix B – Stakeholder Signoff

Stakeholder Name	Site or Department	Contact	Stakeholder Signoff
Francis Drolet	Horne	Francis.Drolet@glencore.ca	DocuSigned by: Francis Drolet 1/12/2021
Charlie Gillmore	Ernest Henry Mining	charlie.gillmore@glencore.com.au	DocuSigned by: Charlie Gillmore 30/11/2021
Neale Laffin	Queensland Metals – Processing and Smelting	neale.laffin@glencore.com.au	DocuSigned by: Neale Laffin 29/11/2021
Camilo Zenis	Lomas Bayas	Camilo.Zenis@glencore.cl	DocuSigned by: Camilo Zenis 1/12/2021
Tony Archer	AM Network	tarcher@amnetwork.com.au	DocuSigned by: Tony Archer 19/11/2021
Akeem Araoye	Asset Management – North America	Akeem.Araoye@glencore.ca	DocuSigned by: Akeem Araoye 19/11/2021
Osvaldo Zamorano	Operational Excellence Metallurgical Assets	Osvaldo.Zamorano@glencore.cl	DocuSigned by: Osvaldo Zamorano 30/11/2021
Martin Samson	Project Services – Metallurgical Assets	Martin.Samson@glencore.ca	DocuSigned by: Martin Samson 29/11/2021
Peter Zuliani	Operational Excellence Metallurgical Assets	Peter.Zuliani@glencore.ca	DocuSigned by: Peter Zuliani 19/11/2021
Dino Moll	Asset Management - Australia	Dino.Moll@glencore.com.au	DocuSigned by: 22/11/2021
Rohan Reisener	Asset Management - Australia	rohan.reisener@glencore.com.au	DocuSigned by: Rohan Reisener 19/11/2021

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Notes

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Shutdowns & Overhauls Management

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